

Why is the QAOA order  $\hat{H} B$  while the

Adiabatic order is  $B \hat{H}$ ?

\* Adiabatic:- we move from say  $|1\rangle$  high energy state of  $X$  then slowly so that the orbit rides the change in Hamiltonian (in this case  $X \rightarrow Z$ ) so when it reaches  $Z$  it is in  $|0\rangle$  while when we abruptly change orbit falls off to  $|1\rangle$  (some superposition like)

\* QAOA:-

\* Consider that input state is a set of arrows  $|\psi\rangle_n$

\* After  $H^{\otimes n}$  all arrows are pointing in

same direction (just basis change) & same amplitude. Think all are pointing North.

\* Now apply  $\hat{H}$  (Hamiltonian) but u hit it with  $e^{-i\gamma\hat{H}}$  which rotates each arrow based on its bitstring "score" (think this just how vector is rotated)

\* So the cut value  $C(x)$  is eigenvalue of  $\hat{H}$



\* as a result what intuitively happens is all good values are in different directions than bad ones!! bcos this  
no amp change! diagonal!

\* Now comes mixer  $e^{-i\beta\hat{B}}$  which makes neighbouring arrows add up tip to tail but  $e^{-i\beta\hat{B}}$  knows nothing about good or bad bitstrings or vectors!!

\* if they point same way then u get longer arrow

\* if they point opposite then u get shorter arrow.

\* As mixer piles neighbours together the reinforcement happens.

\* Now we measure good bitstrings or vectors are more likely!!

\* So amplitude physically flows from bad ones to good ones!!

\* This is QAOA working!!



\* So now order is not negotiable

So it cost for  $\hat{H}$   $\xrightarrow{\text{then}}$  mixer  $\hat{B}$ .

\* If u reverse the  $\hat{B}$  has nothing to add up so ideally nothing should have happened

\* then  $\hat{H}$  comes and would have rotated but no amplitude change so measurement will have no bias!!

\* This means mixer  $\hat{B}$  interference is useful only when  $\hat{H}$  has given the vectors a meaningful direction!!

\* So move to right direction and then combine first!!